# Daily Sequences Drill #3 — Answers & Investigations

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| Section | Topic | Questions | Target time |
|---|---|---|---|
| A | Rapid Recognition | 10 | 2 min 30 sec |
| B | Manipulation Drills | 10 | 8 min |
| C | Substitution & Structure | 8 | 10 min |
| D | Challenge | 5 | 15 min |

*New toolkit today: Geometric Sequences & Series · Sum to Infinity · The Convergence Condition* $|r| < 1$.

## Section A Rapid Recognition

**1.** Write down the common ratio of the GP $3, 6, 12, 24, \ldots$

**Answer:** 2

**Method:** $6/3 = 2$, $12/6 = 2$, $24/12 = 2$ — constant ratio 2.

> **Investigate further:** A GP's ratio is always *next term* ÷ *current term*, never the other way round.

**2.** A GP has first term 5 and common ratio 2. Find the 6th term.

**Answer:** 160

**Method:** $a_6 = 5 \times 2^5 = 5 \times 32 = 160$.

> **Investigate further:** Compare to an AP's $a_n = a + (n-1)d$: a GP's nth term is $a \times r^{n-1}$ — multiplication replaces addition throughout.

**3.** A GP has first term 2 and common ratio 3. Find $S_5$.

**Answer:** 242

**Method:** $S_5 = \dfrac{2(3^5 - 1)}{3 - 1} = \dfrac{2 \times 242}{2} = 242$.

> **Investigate further:** Check by direct addition: $2 + 6 + 18 + 54 + 162 = 242$. ✓

**4.** State the condition on $r$ for an infinite GP's sum to converge.

**Answer:** $|r| < 1$

**Method:** The infinite sum $a + ar + ar^2 + \cdots$ only settles to a finite value when each term shrinks toward 0, which needs $|r| < 1$.

> **Investigate further:** This is the single condition the entire sheet is built around — see Common Traps below for what goes wrong when it's skipped.

**5.** A GP has first term 8 and common ratio $\frac{1}{2}$. Find its sum to infinity.

**Answer:** 16

**Method:** $S_\infty = \dfrac{8}{1 - \frac{1}{2}} = \dfrac{8}{\frac{1}{2}} = 16$.

> **Investigate further:** Sanity check: partial sums $8, 12, 14, 15, 15.5, \ldots$ climb toward 16 but never reach it.

**6.** Which of these three sequences is geometric: (i) $2, 4, 6, 8$ (ii) $2, 4, 8, 16$ (iii) $2, 4, 7, 11$?

**Answer:** (ii) $2, 4, 8, 16$ is geometric (ratio 2).

**Method:** (i) has constant difference 2 — arithmetic, not geometric. (iii) has differences $2, 3, 4$ and ratios $2, 1.75, 1.57\ldots$ — neither.

> **Investigate further:** Always test a sequence against *both* AP and GP definitions before concluding it's neither — (iii) genuinely fits neither pattern here.

**7.** Write down the sum-to-infinity formula for a GP with first term $a$ and ratio $r$ (where $|r| < 1$).

**Answer:** $S_\infty = \dfrac{a}{1-r}$

**Method:** Take the finite-sum formula $S_n = \dfrac{a(1-r^n)}{1-r}$ and let $n \to \infty$: since $|r| < 1$, $r^n \to 0$, leaving $S_\infty = \dfrac{a}{1-r}$.

> **Investigate further:** This derivation — taking a limit of the finite formula — is worth remembering, since it's exactly how D1's proof works too.

**8.** True or false: a GP with common ratio $r = 1$ has a well-defined sum to infinity.

**Answer:** False

**Method:** At $r = 1$ every term equals $a$, so the sum after $n$ terms is $na$, which grows without bound (unless $a = 0$) — it never settles to a finite value.

> **Investigate further:** $r = 1$ is exactly the boundary case excluded by $|r| < 1$ — it's a common slip to think "ratio equals something reasonable" is enough without checking it's strictly less than 1 in magnitude.

**9.** A GP has first term 100 and common ratio $-\frac{1}{2}$. Find its sum to infinity.

**Answer:** $\dfrac{200}{3}$

**Method:** $S_\infty = \dfrac{100}{1-(-\frac{1}{2})} = \dfrac{100}{\frac{3}{2}} = \dfrac{200}{3}$.

> **Investigate further:** Subtracting a negative ratio increases the denominator's effective size compared to a positive ratio of the same magnitude — always expand $1 - (-\frac{1}{2})$ to $1 + \frac{1}{2}$ carefully rather than rushing the sign.

**10.** Write the next term of the GP $81, 27, 9, 3, \ldots$

**Answer:** 1

**Method:** Ratio is $\frac{1}{3}$: $81, 27, 9, 3, 1, \ldots$

> **Investigate further:** This GP's terms shrink toward 0 forever without ever reaching it — exactly the behaviour that makes a sum-to-infinity possible.

## Section B Manipulation Drills

**1.** A GP has first term 4 and common ratio 3. Find the smallest $n$ for which $S_n > 1000$.

**Answer:** $n = 6$

**Method:** $S_n = \dfrac{4(3^n - 1)}{2} = 2(3^n - 1) > 1000 \Rightarrow 3^n > 501$. $3^5 = 243$ (too small), $3^6 = 729 > 501$. So $n = 6$.

> **Investigate further:** Always test the boundary powers directly rather than trying to solve the inequality with logarithms by hand under time pressure — for small bases, a quick doubling/tripling table is faster.

**2.** For which values of $r$ does the infinite GP with first term 5 have a finite sum?

A) all real $r$

B) $r > 0$ only

C) $-1 < r < 1$

D) $r \neq 1$

**Answer:** C) $-1 < r < 1$

**Method:** The sum-to-infinity formula $S_\infty = \dfrac{5}{1 - r}$ is only valid, i.e. only converges, when $|r| < 1$ — the first term's value (5) plays no role in whether convergence happens.

> **Investigate further:** This is worth internalising as a general rule: convergence depends *only* on $r$, never on $a$.

**3.** A GP has first term 3 and common ratio $\frac{2}{3}$. Find its sum to infinity.

**Answer:** 9

**Method:** $S_\infty = \dfrac{3}{1 - \frac{2}{3}} = \dfrac{3}{\frac{1}{3}} = 9$.

> **Investigate further:** Cross-check: the terms are $3, 2, \frac{4}{3}, \frac{8}{9}, \ldots$, and their running total climbs toward 9.

**4.** A GP has sum to infinity 20 and first term 5. Find $r$.

A) $\frac{3}{4}$

B) $\frac{1}{4}$

C) $\frac{4}{5}$

D) $-\frac{3}{4}$

**Answer:** A) $\frac{3}{4}$

**Method:** $20 = \dfrac{5}{1 - r} \Rightarrow 1 - r = \dfrac{5}{20} = \dfrac{1}{4} \Rightarrow r = \dfrac{3}{4}$.

> **Investigate further:** Always isolate $(1 - r)$ first by cross-multiplying, then solve for $r$ last — rearranging in the wrong order is where sign slips creep in.

**5.** A GP has first term 1 and sum to infinity 2. Find the second term.

A) $\frac{1}{2}$

B) $\frac{1}{4}$

C) 2

D) 1

**Answer:** A) $\frac{1}{2}$

**Method:** $2 = \dfrac{1}{1-r} \Rightarrow 1 - r = \dfrac{1}{2} \Rightarrow r = \dfrac{1}{2}$. Second term $= 1 \times \dfrac{1}{2} = \dfrac{1}{2}$.

> **Investigate further:** Don't stop at finding $r$ — the question asks for the second *term*, not the ratio itself, even though here they coincide numerically.

**6.** An AP and a GP both have first term 4. The AP has common difference 2; the GP has common ratio 2. Which sequence has the larger 10th term?

A) The AP

B) The GP

C) They are equal

D) Cannot be determined

**Answer:** B) The GP

**Method:** AP 10th term: $4 + 9(2) = 22$. GP 10th term: $4 \times 2^9 = 4 \times 512 = 2048$. $2048 \gg 22$.

> **Investigate further:** This is the geometric-eventually-beats-arithmetic theme from Day 1's B10, now made explicit and quantitative: by the 10th term the gap is already close to two orders of magnitude.

**7.** A GP has all terms positive and common ratio $r > 1$. Does its sum to infinity exist?

A) Yes, always.

B) No, never — it diverges.

C) Only if the first term is small.

D) Only if $r < 2$.

**Answer:** B) No, never — it diverges.

**Method:** $r > 1$ means $|r| > 1$, which is exactly the case excluded from convergence — each term is strictly bigger than the last, so the partial sums grow without bound.

> **Investigate further:** "All terms positive" is a red herring here — positivity says nothing about convergence; only the size of $|r|$ does.

**8.** Express the recurring decimal $0.\overline{3} = 0.333\ldots$ as a fraction, using the geometric series sum-to-infinity formula.

**Answer:** $\dfrac{1}{3}$

**Method:** $0.\overline{3} = \dfrac{3}{10} + \dfrac{3}{100} + \dfrac{3}{1000} + \cdots$, a GP with $a = \dfrac{3}{10}$, $r = \dfrac{1}{10}$. $S_\infty = \dfrac{3/10}{1-1/10} = \dfrac{3/10}{9/10} = \dfrac{3}{9} = \dfrac{1}{3}$.

> **Investigate further:** Every recurring decimal is secretly a geometric series with $a =$ (the repeating block, as a fraction of its place value) and $r = 10^{-(\text{block length})}$ — C5 generalises this to base 2.

**9.** A GP has first term 2 and common ratio $-3$. Find $S_4$.

A) 40

B) $-40$

C) 160

D) $-160$

**Answer:** B) $-40$

**Method:** $S_4 = \dfrac{2((-3)^4 - 1)}{-3-1} = \dfrac{2(81-1)}{-4} = \dfrac{160}{-4} = -40$.

> **Investigate further:** The trap is A) 40, from mishandling the sign of the denominator $r - 1 = -4$ — always compute $r - 1$ (or $1 - r$, matching whichever form of the formula you're using) as a signed number before dividing.

**10.** Recall from Day 2 that $(1-x)^{-1} = 1 + x + x^2 + \cdots$ for $|x| < 1$. Using $x = \frac{1}{3}$, what does this series sum to?

A) $\frac{3}{2}$

B) $\frac{2}{3}$

C) 3

D) 2

**Answer:** A) $\frac{3}{2}$

**Method:** $(1-x)^{-1} = \dfrac{1}{1-x}$. At $x = \dfrac{1}{3}$: $\dfrac{1}{1-1/3} = \dfrac{1}{2/3} = \dfrac{3}{2}$.

> **Investigate further:** This is literally B3's GP with $a = 1, r = \frac{1}{3}$ in different notation — $(1-x)^{-1}$ at a specific $x$ *is* a sum-to-infinity calculation.

## Section C Substitution & Structure

**1.** A GP has first term 1 and unknown common ratio $r$. Its sum to infinity exists and equals 4. What must be true about $r$?

A) $r = \frac{3}{4}$ exactly.

B) $-1 < r < 1$, but no exact value is determinable.

C) $r = \frac{4}{3}$.

D) Infinitely many values of $r$ satisfy this.

**Answer:** A) $r = \frac{3}{4}$ exactly

**Method:** $\dfrac{1}{1-r} = 4 \Rightarrow 1-r = \dfrac{1}{4} \Rightarrow r = \dfrac{3}{4}$ — a single equation in one unknown has exactly one solution.

> **Investigate further:** The trap (D) imagines the equation is somehow underdetermined — it isn't: with $a$ fixed, $S_\infty = 4$ pins down $r$ uniquely.

**2.** A geometric sequence has first term $a$ and common ratio $r$, both positive integers with $r > 1$. Let $S_n$ denote the sum of the first $n$ terms. Given $S_{18} - S_{12} = kS_6$ for some positive integer $k$, find the smallest possible value of $k$. *(after TMUA 2022 Paper 1 Q8)*

A) $2^6$

B) $2^{12}$

C) $2^{18}$

D) $\dfrac{2^6}{2^6 - 1}$

E) $2^6(2^6 - 1)$

**Answer:** B) $2^{12}$

**Method:** $S_{18} - S_{12} = \dfrac{a(r^{18} - r^{12})}{r-1} = \dfrac{ar^{12}(r^6 - 1)}{r-1} = r^{12} \cdot \dfrac{a(r^6-1)}{r-1} = r^{12}S_6$. So $k = r^{12}$, minimised over integers $r > 1$ at $r = 2$: $k = 2^{12} = 4096$.

> **Investigate further:** The real TMUA question (2022 Paper 1 Q8) uses $S_{30} - S_{20} = kS_{10}$, giving $k = r^{20}$, minimised to $2^{20}$ — same mechanism, the exponent is always (index gap between the two subtracted sums).

**3.** An infinite geometric series has first term $a = 6$ and common ratio $r = \dfrac{3k}{10}$, where $k$ is chosen uniformly at random from the integers $-5, -4, \ldots, 4, 5$ (11 equally likely values). Find the probability that the series converges *and* its sum to infinity exceeds 5. *(after TMUA 2023 Paper 1 Q18)*

A) $\frac{4}{11}$

B) $\frac{5}{11}$

C) $\frac{7}{11}$

D) $\frac{3}{11}$

**Answer:** A) $\frac{4}{11}$

**Method:** Convergence needs $|r| < 1$: $\left|\frac{3k}{10}\right| < 1 \Rightarrow |k| < \frac{10}{3} \approx 3.33 \Rightarrow k \in \{-3, \ldots, 3\}$ (7 of the 11 values). Among all 11 possible $k$, $S_\infty = \dfrac{6}{1 - 3k/10} = \dfrac{60}{10 - 3k} > 5$ requires $60 > 5(10 - 3k) \Rightarrow 60 > 50 - 15k \Rightarrow 15k > -10 \Rightarrow k > -\frac{2}{3} \Rightarrow k \geq 0$ (since $10 - 3k > 0$ throughout the converging range, the inequality direction is preserved). Combined with convergence, valid $k \in \{0, 1, 2, 3\}$ — 4 values. Probability $= \dfrac{4}{11}$.

> **Investigate further:** The real TMUA question (2023 Paper 1 Q18) uses $a = 4$, $r = \frac{2k}{7}$, threshold $S > 3$, landing on $\frac{5}{11}$ — the denominator is always the *full* sample space (11), not just the converging subset, because a non-converging $k$ automatically fails the "sum exceeds the threshold" condition rather than being excluded from the count.

**4.** The series $\sum_{k=1}^{\infty} ar^{k-1}$ converges (for real $a, r$) when $|r| < 1$. If $a$ and $r$ are instead allowed to be complex numbers, which is the correct convergence condition?

A) $|r| < 1$ (the same condition, using the complex magnitude).

B) $r < 1$ (a real inequality).

C) $a < 1$.

D) There is no meaningful convergence condition for complex $r$.

**Answer:** A) $|r| < 1$ (using the complex magnitude)

**Method:** The geometric series' convergence proof relies only on $|r^n| \to 0$, and $|r^n| = |r|^n$ holds identically whether $r$ is real or complex — so the same magnitude condition transfers directly.

> **Investigate further:** This is a genuine abstraction check: convergence of a geometric series is fundamentally about the *size* of $r$ shrinking powers to 0, a concept that doesn't care whether $r$ lives on the real line or in the complex plane.

**5.** Convert the recurring base-2 number $0.\overline{0111} = 0.0111\,0111\,0111\ldots$ (base 2) to a fraction in base 10. *(after TMUA 2023 Paper 2 Q15)*

A) $\frac{1}{3}$

B) $\frac{1}{5}$

C) $\frac{7}{15}$

D) $\frac{7}{16}$

**Answer:** C) $\dfrac{7}{15}$

**Method:** The repeating block 0111 (base 2) has integer value $0 \times 8 + 1 \times 4 + 1 \times 2 + 1 \times 1 = 7$, over a 4-bit block. This is a GP with $a = \dfrac{7}{16}$, $r = \dfrac{1}{16}$: $S_\infty = \dfrac{7/16}{1 - 1/16} = \dfrac{7/16}{15/16} = \dfrac{7}{15}$.

> **Investigate further:** The real TMUA question (2023 Paper 2 Q15) uses the block 0011 (value 3), giving $\dfrac{3}{15} = \dfrac{1}{5}$ — same mechanism as B8's base-10 recurring decimal, just with block value and place value both computed in base 2 instead of base 10.

**6.** Which of the following statements about arithmetic and geometric series is FALSE?

A) An AP's terms grow linearly in $n$; a GP's terms with $|r| > 1$ grow exponentially in $n$.

B) An AP diverges to $\pm\infty$ (or stays constant) as $n \to \infty$, unless $d = 0$.

C) A GP can converge to a finite sum to infinity even though no individual term is exactly zero, provided $|r| < 1$.

D) Every convergent GP has all its terms equal to zero eventually.

**Answer:** D) Every convergent GP has all its terms equal to zero eventually.

**Method:** A convergent GP's terms $ar^{n-1} \to 0$ as a *limit*, but for any finite $n$ and $r \neq 0$, $ar^{n-1} \neq 0$ exactly — the terms get arbitrarily small without ever reaching zero. A), B), C) are all genuinely true statements.

> **Investigate further:** This is the same "limit vs. actually reaching a value" idea as D2's $0.\overline{9} = 1$ — a convergent sequence's terms approach 0, and its partial sums approach $S_\infty$, but neither is literally attained at any finite step.

**7.** Using $(1-x)^{-1} = \sum_{k=0}^{\infty} x^k$ from Day 2, evaluate $\displaystyle\sum_{k=0}^{\infty} \frac{1}{5^k}$ by choosing an appropriate value of $x$.

A) $\frac{5}{4}$

B) $\frac{4}{5}$

C) 5

D) $\frac{1}{4}$

**Answer:** A) $\frac{5}{4}$

**Method:** $\displaystyle\sum_{k=0}^{\infty} \frac{1}{5^k} = \sum_{k=0}^{\infty} \left(\frac{1}{5}\right)^k = (1-\tfrac{1}{5})^{-1}$ at $x = \frac{1}{5}$: $(1-\frac{1}{5})^{-1} = (\frac{4}{5})^{-1} = \dfrac{5}{4}$.

> **Investigate further:** Recognising $\sum \frac{1}{5^k}$ as $\sum x^k$ with $x = \frac{1}{5}$ turns a series-summing problem into a one-line substitution — exactly Day 2's payoff.

**8.** Let $S = 1+2x+3x^2+4x^3+\cdots$. Recall from Day 2 that this is exactly $(1-x)^{-2} = \sum_{k=0}^{\infty}(k+1)x^k$. Evaluate $S$ at $x = \frac{1}{4}$.

A) $\frac{16}{9}$

B) $\frac{4}{3}$

C) $\frac{9}{16}$

D) $\frac{3}{4}$

**Answer:** A) $\frac{16}{9}$

**Method:** $(1-x)^{-2}$ at $x = \frac{1}{4}$: $(1-\frac{1}{4})^{-2} = (\frac{3}{4})^{-2} = \left(\frac{4}{3}\right)^2 = \dfrac{16}{9}$.

> **Investigate further:** This series $1 + 2x + 3x^2 + \cdots$ is literally the derivative of $1 + x + x^2 + \cdots$ term-by-term (each $x^k$ becomes $kx^{k-1}$, then re-indexed) — Day 2's D3 proved the finite analogue of exactly this differentiation trick.

## Section D Challenge

**1.** Prove that $\displaystyle\sum_{k=1}^{\infty} \frac{1}{2^k} = 1$ exactly (not merely "approaches 1"), using the sum-to-infinity formula.

Then explain why the partial sum $S_n = \sum_{k=1}^{n} \dfrac{1}{2^k} = 1 - \dfrac{1}{2^n}$ never actually equals 1 for any finite $n$.

**Answer:** Proof via the sum-to-infinity formula; partial sums approach but never reach 1.

**Method:** $a = \frac{1}{2}, r = \frac{1}{2}$: $S_\infty = \dfrac{1/2}{1-1/2} = \dfrac{1/2}{1/2} = 1$ exactly, by the formula itself (not an approximation).

Separately, $S_n = \dfrac{a(1-r^n)}{1-r} = \dfrac{\frac{1}{2}(1-(\frac{1}{2})^n)}{\frac{1}{2}} = 1 - \dfrac{1}{2^n}$, which is strictly less than 1 for every finite $n$ (since $\frac{1}{2^n} > 0$ always) — the infinite sum is the *limit* of these partial sums, not a partial sum itself.

> **Investigate further:** This is the precise resolution of the "does 0.999... really equal 1" confusion in D2: the infinite sum is defined as exactly the limit, and the limit is exactly 1, even though no finite partial sum ever is.

**2.** Let $x = 0.\overline{9} = 0.999\ldots$ (a recurring 9). Writing $x$ as an infinite geometric series and summing it, determine $x$ exactly.

A) $x = 1$ exactly.

B) $x$ is infinitesimally less than 1 but not equal to it.

C) $0.\overline{9}$ is undefined.

D) $x = \frac{9}{10}$.

**Answer:** A) $x = 1$ exactly.

**Method:** $0.\overline{9} = \dfrac{9}{10} + \dfrac{9}{100} + \cdots$, a GP with $a = \dfrac{9}{10}, r = \dfrac{1}{10}$. $S_\infty = \dfrac{9/10}{1-1/10} = \dfrac{9/10}{9/10} = 1$.

> **Investigate further:** Exactly D1's mechanism, applied to a famous "paradox" — there is no meaningful sense in which $0.\overline{9}$ is "slightly less than 1"; it is defined as the limit of its partial sums, and that limit is exactly 1.

**3.** A ball is dropped from height $h$. Each time it bounces, it rebounds to a fraction $r$ of its previous height ($0 < r < 1$). Find, in terms of $h$ and $r$, the total distance the ball travels (up and down, forever) before coming to rest.

A) $h \cdot \dfrac{1+r}{1-r}$

B) $h \cdot \dfrac{2}{1-r}$

C) $h \cdot \dfrac{1}{1-r}$

D) $h(1+2r)$

**Answer:** A) $h \cdot \dfrac{1+r}{1-r}$

**Method:** Total distance = (initial drop $h$) + (every rebound-and-fall pair, each pair covering $2\times$ that rebound's height). Rebound heights are $hr, hr^2, hr^3, \ldots$, a GP with first term $hr$, ratio $r$. Distance from rebounds $= 2 \times \dfrac{hr}{1-r}$. Total $= h + \dfrac{2hr}{1-r} = \dfrac{h(1-r)+2hr}{1-r} = \dfrac{h(1+r)}{1-r}$.

> **Investigate further:** The factor of 2 per rebound (up then down) is the easiest part to forget — always separate the one unpaired initial drop from the infinitely many paired bounces before summing.

**4.** The sum to infinity of a GP equals the sum of its first two terms. What can be said about the common ratio $r$?

A) $r = 0$ (the "GP" is actually constant).

B) $r = \pm 1$.

C) $r = \frac{1}{2}$.

D) No such $r$ exists.

**Answer:** A) $r = 0$

**Method:** $S_\infty = a(1+r) \Rightarrow \dfrac{a}{1-r} = a(1+r) \Rightarrow \dfrac{1}{1-r} = 1+r$ (for $a \neq 0$) $\Rightarrow 1 = (1+r)(1-r) = 1 - r^2 \Rightarrow r^2 = 0 \Rightarrow r = 0$.

> **Investigate further:** $r = 0$ makes every term after the first equal to 0, so the "GP" is really just the constant $a$ — a degenerate edge case, but the algebra forces it uniquely rather than allowing it as one option among several.

**5.** A sequence is defined by $a_1 = 3$ and $a_{n+1} = \frac{1}{2}a_n$ for $n \geq 1$. Using the sum-to-infinity formula, evaluate $\displaystyle\sum_{n=1}^{\infty} a_n$.

A) 6

B) 3

C) 4.5

D) It cannot be summed, since it's defined by a recurrence, not a formula.

**Answer:** A) 6

**Method:** The recurrence $a_{n+1} = \frac{1}{2}a_n$ with $a_1 = 3$ generates exactly the GP $3, \frac{3}{2}, \frac{3}{4}, \ldots$ (each term is half the last, matching $r = \frac{1}{2}$). $S_\infty = \dfrac{3}{1-\frac{1}{2}} = 6$.

> **Investigate further:** This is worth noticing directly: a recurrence relation defined by "multiply by a constant each time" is always secretly a GP in disguise, solvable with today's toolkit rather than needing anything new — Day 4 tackles recurrences that *aren't* this simple.

## Top 5 Patterns — Worksheet #3

1. **Sum to infinity: $S_\infty = \dfrac{a}{1-r}$, valid only for $|r| < 1$.** The central formula of the whole sheet. *(see A5, A7, A9, B3, B4)*

2. **Always check $|r| < 1$ before writing down $S_\infty$ at all.** Convergence depends only on $r$, never on $a$. *(see A4, A8, B2, B7)*

3. **Recurring decimals are geometric series in disguise.** $a =$ the repeating block's value (as a fraction of its place value), $r = (\text{base})^{-(\text{block length})}$. *(see B8, C5, D2)*

4. **Day 2's $(1-x)^{-1} = \sum x^k$ *is* the geometric series.** Plug in any $|x| < 1$ to evaluate a numeric infinite series instantly. *(see B10, C7, C8)*

5. **Know which formula a question wants: finite $S_n = \dfrac{a(r^n - 1)}{r - 1}$ or infinite $S_\infty = \dfrac{a}{1-r}$.** They answer different questions and only one needs a convergence check. *(see A3, B1, B9, C2)*

## Common Traps — Worksheet #3

1. **Applying $S_\infty = \frac{a}{1-r}$ without checking $|r| < 1$ first.** The formula still spits out a number even when the series actually diverges — that number is meaningless. *(see B7, C1)*

2. **Sign errors with negative common ratios inside $1 - r$.** $1 - (-\frac{1}{2}) = \frac{3}{2}$, not $\frac{1}{2}$ — expand the double negative carefully. *(see A9, B9)*

3. **Confusing "converges" with "sums to something small."** A GP with $r$ just under 1 still converges, but its sum can be enormous. *(see C6)*

4. **Treating a probability question's sample space as only the converging outcomes.** Non-convergent cases stay in the denominator — they just automatically fail the sum condition, they aren't removed from the count. *(see C3)*

5. **Assuming every recurrence relation needs unfamiliar machinery.** Some recurrences ("multiply by a constant each step") are secretly a GP, solvable with this sheet's tools already. *(see D5)*

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*"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."*

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Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths